Quasi-static viscoelastic and elasto-plastic solids on Jexpresso's spectral elements

The thick-walled cylinder of Artioli, Beirão da Veiga, Lovadina & Sacco (2017), §4.2

Jexpresso, branch `sm/solid`, commit `ff10c6c`

Abstract

This note documents the quasi-static solid-mechanics path of Jexpresso (`:lquasistatic_solid`) and the benchmark it was built for: a plane-strain thick-walled cylinder under a step internal pressure, made of a generalized Maxwell viscoelastic material (Artioli et al. 2017, Secs. 3.1 and 4.2). It states the governing equations, the constitutive laws (the viscoelastic one of the benchmark and the J_2 plasticity that uses the same quadrature-point history), every step of the spectral-element discretisation, and the initialisation, and links each to the line of the Jexpresso source where it is done. Every link is a permalink to commit `ff10c6c`.

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1 The problem

A long cylinder of inner radius $a = 2$ and outer radius $b = 4$ is loaded by an internal pressure $p = 10$ switched on at $t = 0$ and held to $t = 20$; the outer surface is free. The cylinder is long, so the state is *plane strain* ($u_z = 0$, $\varepsilon_{zz} = 0$), and by symmetry one quarter of the cross-section is modelled, with rollers on the two straight edges (Fig. 1). The control points are $A = (a, 0)$ on the bore and $B = (b, 0)$ on the outer surface, where $u_r = u_x$.

Code: the case: `problems/Elasticity/viscoelastic_cylinder2d`; deck `user_inputs`; material, geometry, load `cylinder_properties`; mesh `structured_quarter_annulus`.

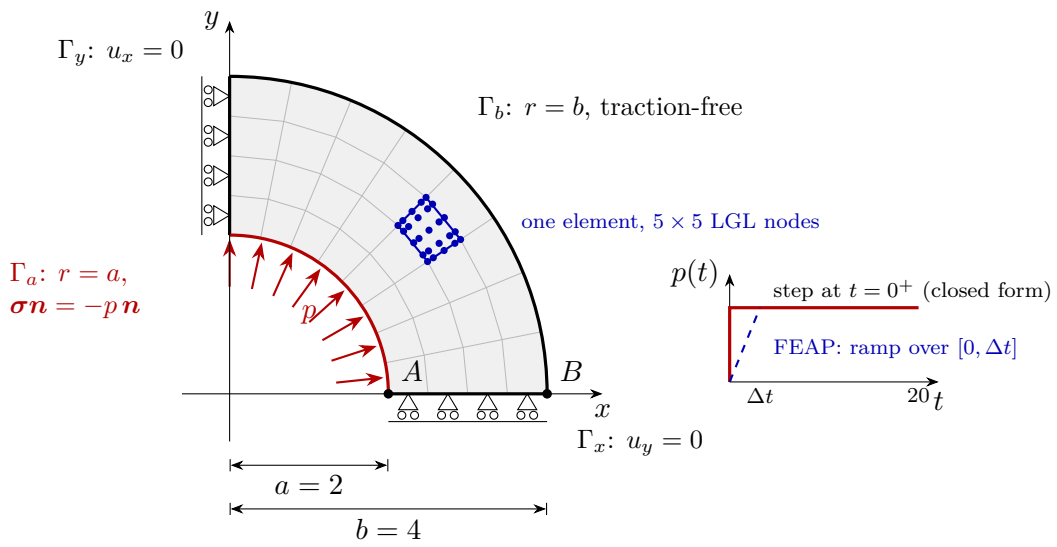


Figure 1: The quarter cylinder (plane strain) and its boundary conditions, drawn over the spectral-element grid of the case: 4×8 quadrilaterals, both arcs curved onto their circles; one element shows its 5×5 LGL nodes ($N = 4$). Inset: the pressure history — the Heaviside step of the closed form, and the ramp over the first increment that the paper's FEAP runs amount to (§5).

quantity	value	note
Young's modulus, Poisson's ratio	$E = 1000$, $\nu = 0.3$	instantaneous
shear, bulk modulus	$G = E/2(1+\nu) = 384.6$, $K = E/3(1-2\nu) = 833.3$	
Prony series	$G(t) = G\,(\mu_0 + \mu_1 e^{-t/\lambda})$, $\lambda = 1$	λ: relaxation time
set ve1 / ve2	$(\mu_0, \mu_1) = (0.01, 0.99)$ / $(0.3, 0.7)$	`JEXPRESSO_VE_SET`
geometry, load	$a = 2$, $b = 4$, $p = 10$, $t \in [0, 20]$	
discretisation	$N = 4$, 4×8 elements, $\Delta t = 1$	1122 DOFs

Table 1: Data of the benchmark. Note: the paper's λ is a *relaxation time*; it has nothing to do with the Lamé constant, which is why the code calls it τ.

2 Governing equations

Small strains, no inertia (the loading is slow compared with the elastic wave transit, so the problem is *quasi-static*), no body force. For every $t \in [0, T]$ find the displacement $\boldsymbol{u}(\boldsymbol{x}, t)$ on the quarter cross-section Ω such that

$$\nabla\cdot\boldsymbol{\sigma} + \boldsymbol{b} = \boldsymbol{0} \qquad \text{in } \Omega, \tag{1}$$

$$\boldsymbol{\varepsilon}(\boldsymbol{u}) = \operatorname{sym}\nabla\boldsymbol{u} = \tfrac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}}), \quad \varepsilon_{zz} = 0 \qquad \text{(plane strain)}, \tag{2}$$

$$\boldsymbol{\sigma n} = -p(t)\,\boldsymbol{n} \qquad \text{on } \Gamma_a \ (r = a), \tag{3}$$

$$\boldsymbol{\sigma n} = \boldsymbol{0} \qquad \text{on } \Gamma_b \ (r = b), \tag{4}$$

$$u_y = 0, \quad (\boldsymbol{\sigma n})_x = 0 \qquad \text{on } \Gamma_x \ (y = 0), \tag{5}$$

$$u_x = 0, \quad (\boldsymbol{\sigma n})_y = 0 \qquad \text{on } \Gamma_y \ (x = 0), \tag{6}$$

with $\boldsymbol{n}$ the outward unit normal and $\boldsymbol{\sigma}$ given by the constitutive law of §3, which carries the memory of the material. Time enters *only* through that memory and through $p(t)$.

Weak form. With $V = \{\boldsymbol{v} \in H^1(\Omega)^2 : v_y = 0 \text{ on } \Gamma_x,\ v_x = 0 \text{ on } \Gamma_y\}$, find $\boldsymbol{u}(t) \in V$ such that

$$\int_\Omega \boldsymbol{\varepsilon}(\boldsymbol{v}) : \boldsymbol{\sigma}\, \mathrm{d}\Omega = \int_{\Gamma_a} \boldsymbol{v} \cdot (-p\,\boldsymbol{n})\, \mathrm{d}\Gamma + \int_\Omega \boldsymbol{v} \cdot \boldsymbol{b}\, \mathrm{d}\Omega \qquad \forall\, \boldsymbol{v} \in V. \tag{7}$$

Only the in-plane components of $\boldsymbol{\sigma}$ appear ($\varepsilon_{zz}(\boldsymbol{v}) = 0$), but the material point is three-dimensional and σ_{zz} is part of the solution.

Code: boundary conditions `user_solid_bc`; body force `user_solid_body_force`; the solver and its documentation `quasistatic_solid.jl`; dispatch from the driver `drivers.jl`.

3 Constitutive laws

3.1 Generalized Maxwell viscoelasticity (the benchmark)

Volumetric–deviatoric split, $\boldsymbol{\varepsilon} = \boldsymbol{e} + \frac{1}{3}\operatorname{tr}(\boldsymbol{\varepsilon})\mathbf{1}$, elastic in volume and generalized Maxwell in the deviator (Artioli et al. eqs. 3.2–3.7):

$$\boldsymbol{\sigma} = K\ \operatorname{tr}(\boldsymbol{\varepsilon})\,\mathbf{1} + \boldsymbol{s}, \qquad \boldsymbol{s}(t) = \int_{-\infty}^{t} 2G(t-s)\,\dot{\boldsymbol{e}}(s)\,\mathrm{d}s, \qquad G(t) = G\Big(\mu_0 + \sum_{m=1}^{M} \mu_m e^{-t/\lambda_m}\Big), \tag{8}$$

with $\mu_0 + \sum_m \mu_m = 1$. Each Prony term is a spring–dashpot branch whose spring strain $\boldsymbol{q}_m$ (a deviatoric tensor) obeys

$$\dot{\boldsymbol{q}}_m + \frac{1}{\lambda_m}\boldsymbol{q}_m = \dot{\boldsymbol{e}}, \qquad \boldsymbol{s} = 2G\Big(\mu_0\,\boldsymbol{e} + \sum_m \mu_m \boldsymbol{q}_m\Big). \tag{9}$$

In plane strain $e_{zz} = -\frac{1}{3}\operatorname{tr}\boldsymbol{\varepsilon} \neq 0$, so $\boldsymbol{q}_m$ and $\boldsymbol{s}$ have four non-zero components (xx, yy, zz, xy) and $\sigma_{zz} = K\operatorname{tr}\boldsymbol{\varepsilon} + s_{zz}$ creeps.

Time integration: the exponential recursion. Integrating (9) exactly over $[t_n, t_{n+1}]$ for a strain that is linear in time (Zienkiewicz, Taylor & Fox 2014; Simo & Hughes 1998, §10.3) gives

$$\boldsymbol{q}_m^{n+1} = d_m\,\boldsymbol{q}_m^{n} + \gamma_m(\boldsymbol{e}^{n+1} - \boldsymbol{e}^{n}), \qquad d_m = e^{-\Delta t/\lambda_m}, \quad \gamma_m = \frac{\lambda_m}{\Delta t}(1 - e^{-\Delta t/\lambda_m}), \tag{10}$$

second-order accurate; it is the update FEAP — and so the paper — uses. Simo & Hughes' Box 10.1 variant $\gamma_m = e^{-\Delta t/2\lambda_m}$ is also second order; backward Euler $d_m = \gamma_m = (1 + \Delta t/\lambda_m)^{-1}$ is first order. Because (10) is linear in $\boldsymbol{e}^{n+1}$,

$$\boldsymbol{\sigma}^{n+1} = K\operatorname{tr}(\boldsymbol{\varepsilon}^{n+1})\mathbf{1} + 2\tilde{G}\,\boldsymbol{e}^{n+1} + \boldsymbol{\sigma}^{*}, \qquad \tilde{G} = G\Big(\mu_0 + \sum_m \mu_m\gamma_m\Big), \quad \boldsymbol{\sigma}^{*} = 2G\sum_m \mu_m(d_m\boldsymbol{q}_m^{n} - \gamma_m\boldsymbol{e}^{n}) : \tag{11}$$

each step is a linear elastic problem with the *algorithmic* shear modulus $\tilde{G}$ and a known initial stress $\boldsymbol{\sigma}^{*}$. For $\Delta t = 0$, $d_m = \gamma_m = 1$: the instantaneous (glassy) response.

Code: coefficients (d_m, γ_m) `maxwell_step_coefficients`; the material point `solid_constitutive_update!`; the case's material `user_solid_material`.

Closed-form solution. By the correspondence principle (load p/s, $G \to s\hat{G}(s)$, K unchanged), for the Heaviside pressure

$$u_r(r,t) = P\Big[r\,C(t) + \frac{b^2}{2r}J(t)\Big], \qquad P = \frac{p\,a^2}{b^2-a^2}, \tag{12}$$

$$J(t) = \frac{1}{G}\Big[\frac{1}{\mu_0} - \Big(\frac{1}{\mu_0} - 1\Big)e^{-\mu_0 t/\lambda}\Big] \quad \text{(shear creep compliance)}, \tag{13}$$

$$C(t) = \frac{1}{M_\infty} - \Big(\frac{1}{M_\infty} - \frac{1}{M_0}\Big)e^{-M_\infty t/(M_0\lambda)}, \qquad M_0 = 2K + \tfrac{2}{3}G, \quad M_\infty = 2K + \tfrac{2}{3}\mu_0 G. \tag{14}$$

At $t = 0$ it is the Lamé solution $P\,\frac{1+\nu}{E}\big[(1-2\nu)r + b^2/r\big]$. $\sigma_{rr} = P(1 - b^2/r^2)$ and $\sigma_{\theta\theta} = P(1 + b^2/r^2)$ do not depend on the material (both surfaces carry tractions), so they are the Lamé stresses at every t; what creeps is u_r and $\sigma_{zz}(t) = 2P\big[\nu_\infty + (\nu - \nu_\infty)e^{-t/\tau_z}\big]$, with $\nu_\infty = \frac{3K - 2\mu_0 G}{2(3K + \mu_0 G)}$, $\tau_z = \lambda\frac{3K+G}{3K+\mu_0 G}$.

Code: `cylinder_ur_exact`, `cylinder_szz_exact`, and the exact-in-space time-discrete reference `cylinder_time_discrete_reference`.

3.2 J_2 plasticity: the history at the quadrature points

The same path carries models whose history is *not* a linear function of the displacement, and stores it at every quadrature point. Von Mises plasticity with linear isotropic and kinematic hardening (Artioli et al. §3.2; Simo & Hughes Box 3.2):

$$\boldsymbol{s} = 2G(\boldsymbol{e} - \boldsymbol{e}^p), \quad \boldsymbol{\alpha} = H_{\text{kin}}\,\boldsymbol{e}^p, \quad \boldsymbol{X} = \boldsymbol{s} - \boldsymbol{\alpha}, \quad f = \|\boldsymbol{X}\| - \sqrt{\tfrac{2}{3}}(\sigma_{y0} + H_{\text{iso}}\,\bar{e}^p) \le 0, \tag{15}$$

$$\dot{\boldsymbol{e}}^p = \dot{\zeta}\,\frac{\boldsymbol{X}}{\|\boldsymbol{X}\|}, \qquad \dot{\bar{e}}^p = \sqrt{\tfrac{2}{3}}\,\dot{\zeta}, \qquad \dot{\zeta} \ge 0,\ \dot{\zeta} f = 0. \tag{16}$$

History per point: $\boldsymbol{e}^p$ (four components) and $\bar{e}^p$. Backward Euler gives the radial return: trial $\boldsymbol{X}^{\text{tr}} = 2G(\boldsymbol{e}^{n+1} - \boldsymbol{e}^{p,n}) - H_{\text{kin}}\boldsymbol{e}^{p,n}$; if $f^{\text{tr}} > 0$, $\Delta\zeta = f^{\text{tr}}/(2G + H_{\text{kin}} + \frac{2}{3}H_{\text{iso}})$, $\boldsymbol{n} = \boldsymbol{X}^{\text{tr}}/\|\boldsymbol{X}^{\text{tr}}\|$, $\boldsymbol{e}^{p,n+1} = \boldsymbol{e}^{p,n} + \Delta\zeta\boldsymbol{n}$, and the consistent tangent is

$$\mathbb{C} = K\,\mathbf{1}\otimes\mathbf{1} + 2G\theta\,\mathbb{I}_{\text{dev}} - 2G\bar{\theta}\,\boldsymbol{n}\otimes\boldsymbol{n}, \quad \theta = 1 - \frac{2G\Delta\zeta}{\|\boldsymbol{X}^{\text{tr}}\|}, \quad \bar{\theta} = \frac{2G}{2G + H_{\text{kin}} + \frac{2}{3}H_{\text{iso}}} - (1-\theta). \tag{17}$$

A shape-memory alloy (Artioli et al. §3.3) keeps the transformation strain $\boldsymbol{e}^{tr}$ where J_2 keeps $\boldsymbol{e}^p$: the storage, the Newton loop of Step 7 (§4) and the output are the same.

Code: `solid_constitutive_update!` (J_2); history layout and plottable fields `solid_history_fields`; the plasticity case `problems/Elasticity/plastic_cylinder2d`.

4 Spectral-element discretisation, step by step

Every spatial operator below is Jexpresso's own; the solver adds the material and the time stepping.

Step 1 — the grid

gmsh writes a structured, *straight-sided* grid of 4×8 quadrilaterals with the physical groups `inner`, `outer`, `xaxis`, `yaxis` and `domain`. Jexpresso reads it, places on each element the $(N+1)^2$ Legendre–Gauss–Lobatto (LGL) nodes $\boldsymbol{\xi}_{ij} = (\xi_i, \xi_j)$ of degree $N = 4$, and numbers the nodes globally through the connectivity $\text{conn}(e, i, j)$ (`connijk`). gmsh leaves every high-order node of an arc on its

chord; `:exact_geometry` snaps the boundary nodes onto the circles $r = a$ and $r = b$ and blends the correction into the boundary elements (Gordon–Hall), so the boundary is the isoparametric degree-N interpolant of the circle.

Code: `make_meshes.py`; `mod_mesh_read_gmsh!`; `exact_geometry.jl` (called at `mesh.jl`); deck `:exact_geometry`.

Step 2 — basis, quadrature and metric terms

On the reference square the basis is the tensor product of the degree-N Lagrange polynomials through the LGL nodes, $\psi_{ij}(\xi,\eta) = \ell_i(\xi)\ell_j(\eta)$, and the displacement is

$$\boldsymbol{u}_h(\boldsymbol{x}) = \sum_a \psi_a(\boldsymbol{x})\,\boldsymbol{u}_a, \qquad a = \mathrm{conn}(e,i,j). \tag{18}$$

Integrals use the LGL rule on the same nodes (collocation, $Q = N$; Jexpresso's `Inexact` quadrature), weights $\omega_k\omega_l$. The map $\boldsymbol{x}(\xi,\eta) = \sum \psi_{ij}\boldsymbol{x}_{ij}$ gives the metric terms $\partial\xi/\partial x, \ldots$ and J at every node; with them

$$\frac{\partial\psi_{ij}}{\partial x}\Big|_{kl} = \ell_i'(\xi_k)\ell_j(\xi_l)\,\xi_x + \ell_i(\xi_k)\ell_j'(\xi_l)\,\eta_x, \qquad \int_{\Omega_e} g\,\mathrm{d}\Omega \approx \sum_{k,l}\omega_k\omega_l\,J_{kl}\,g_{kl}. \tag{19}$$

On a boundary edge, the surface Jacobian $J^f = |\mathrm{d}\boldsymbol{x}/\mathrm{d}\xi|$ and the outward normal come from the tangent of the same degree-N interpolant, so they are right on the curved arcs.

Code: `sem_setup`; `build_metric_terms!` (2D, with the boundary normals and J^f at `metric_terms.jl`).

Step 3 — element matrices and direct stiffness summation

Inserting $\boldsymbol{u}_h$ and $\boldsymbol{v} = \psi_a\boldsymbol{e}_\alpha$ into (7) with the Voigt form $\boldsymbol{\sigma} = \mathbf{D}\,\boldsymbol{\varepsilon}$ ($\boldsymbol{\varepsilon} = (\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})$, $\gamma_{xy} = 2\varepsilon_{xy}$, $\mathbf{B}_a = [\partial_x\psi_a\ 0;\ 0\ \partial_y\psi_a;\ \partial_y\psi_a\ \partial_x\psi_a]$) gives the stiffness $\mathbf{K}_{ab} = \sum_e \int_{\Omega_e} \mathbf{B}_a^{\mathsf{T}}\mathbf{D}\,\mathbf{B}_b\,\mathrm{d}\Omega$. Jexpresso builds it from the directional element matrices

$$L_e^{\alpha\beta}[a,b] = \int_{\Omega_e} \partial_\alpha\psi_a\,\partial_\beta\psi_b\,\mathrm{d}\Omega \approx \sum_{k,l}\omega_k\omega_l J_{kl}\,\partial_\alpha\psi_a\partial_\beta\psi_b\big|_{kl}, \tag{20}$$

the components of Jexpresso's Laplacian element matrix ($L_e^{xx} + L_e^{yy}$ is exactly it), assembled by direct stiffness summation (DSS) into sparse global matrices $S_{\alpha\beta} = \mathrm{DSS}(L^{\alpha\beta})$. For the isotropic material, $\mathbf{D} = K\mathbf{D}_{\mathrm{vol}} + \tilde{G}\mathbf{D}_{\mathrm{dev}}$ with $\mathbf{D}_{\mathrm{vol}} = \left[\begin{smallmatrix}1&1&0\\1&1&0\\0&0&0\end{smallmatrix}\right]$, $\mathbf{D}_{\mathrm{dev}} = \left[\begin{smallmatrix}4/3&-2/3&0\\-2/3&4/3&0\\0&0&1\end{smallmatrix}\right]$, so in the flat layout $\mathbf{u} = [\mathbf{u}_x;\mathbf{u}_y]$

$$\mathbf{K}(\tilde{G}) = K\,\mathbf{A}_{\mathrm{vol}} + \tilde{G}\,\mathbf{A}_{\mathrm{dev}}, \quad \mathbf{A}_{\mathrm{vol}} = \begin{bmatrix} S_{xx} & S_{xy} \\ S_{yx} & S_{yy}\end{bmatrix}, \quad \mathbf{A}_{\mathrm{dev}} = \begin{bmatrix} \frac{4}{3}S_{xx} + S_{yy} & S_{yx} - \frac{2}{3}S_{xy} \\ S_{xy} - \frac{2}{3}S_{yx} & S_{xx} + \frac{4}{3}S_{yy}\end{bmatrix}, \tag{21}$$

$S_{yx} = S_{xy}^{\mathsf{T}}$, assembled once. When the tangent varies from point to point (plasticity) the four displacement blocks are built with $\mathbf{D}_{kl}$ inside the quadrature sum and assembled the same way.

Code: `build_laplace_matrix` and its components `build_gradient_product_matrices`; variable tangent `build_tangent_stiffness_matrices`; DSS `DSS_laplace_sparse`; assembly of (21) `solid_sem_operators`, `solid_tangent_matrix`.

Step 4 — the load vector

The pressure (3) is integrated on every boundary edge of Γ_a with the LGL rule of the edge and the curved-edge Jacobian, then summed onto the nodes:

$$\mathbf{f}_a = \sum_{\text{edges}} \sum_{k} \omega_k \, J_k^f \, (-p\,\boldsymbol{n}_k)\, \delta_{a,\text{node}(k)}. \tag{22}$$

A body force would enter through the lumped (diagonal) LGL mass matrix, $\mathbf{f}_a \mathrel{+}= M_a \boldsymbol{b}(\boldsymbol{x}_a)$.

Code: `solid_boundary_conditions!` with Jexpresso's `compute_segment_integral!` and `DSS_segment_integral!`; body force `solid_add_body_force!`.

Step 5 — the symmetry conditions

Each roller fixes one Cartesian component ($u_y = 0$ on Γ_x, $u_x = 0$ on Γ_y); together they remove both translations and the rotation. The constrained DOFs c are eliminated and the solve is on the free block f, which stays symmetric positive definite:

$$\mathbf{K}_{ff}\,\mathbf{u}_f = \mathbf{f}_f - \mathbf{K}_{fc}\,\bar{\mathbf{u}}_c, \qquad \mathbf{u}_c = \bar{\mathbf{u}}_c. \tag{23}$$

$\mathbf{K}_{ff}$ depends on Δt only through $\tilde{G}$, so one sparse Cholesky factorisation serves every step (a second one the instantaneous step).

Step 6 — marching in time: the history as a nodal field

The discrete strain is linear in the nodal displacements and (10) is linear and starts from zero, so at every quadrature point $\boldsymbol{q}_m = \operatorname{dev}\boldsymbol{\varepsilon}(\mathbf{w}_m)$ of a *nodal* field $\mathbf{w}_m$ obeying the same recursion. With $\mathbf{v}\cdot\mathbf{A}_{\text{dev}}\mathbf{z} = \int \boldsymbol{\varepsilon}(\mathbf{v}) : 2\operatorname{dev}\boldsymbol{\varepsilon}(\mathbf{z})$, (11) becomes, for every step $n \to n+1$,

$$\begin{aligned} (K\mathbf{A}_{\text{vol}} + \tilde{G}\mathbf{A}_{\text{dev}})\mathbf{u}^{n+1} &= \mathbf{f}(t_{n+1}) - G\,\mathbf{A}_{\text{dev}} \sum_m \mu_m (d_m \mathbf{w}_m^n - \gamma_m \mathbf{u}^n), \\ \mathbf{w}_m^{n+1} &= d_m \mathbf{w}_m^n + \gamma_m(\mathbf{u}^{n+1} - \mathbf{u}^n), \end{aligned} \tag{24}$$

which reproduces the quadrature-point recursion to round-off.

Algorithm 1 One run of the viscoelastic cylinder (nodal history)

1: assemble $\mathbf{A}_{\text{vol}}, \mathbf{A}_{\text{dev}}$ (Step 3); $\mathbf{u} \leftarrow \mathbf{0}$, $\mathbf{w}_m \leftarrow \mathbf{0}$
2: **if** loads on at t_0: one step of (24) with $\Delta t = 0$ ($d = \gamma = 1$, $\tilde{G} = G$) ▷ glassy response
3: **for** $n = 0, 1, \ldots$ **do**
4: $\quad d_m, \gamma_m \leftarrow$ (10); $\quad \tilde{G} \leftarrow G(\mu_0 + \sum \mu_m \gamma_m)$
5: $\quad \mathbf{f} \leftarrow$ pressure load (Step 4) $- G\mathbf{A}_{\text{dev}} \sum_m \mu_m (d_m \mathbf{w}_m - \gamma_m \mathbf{u})$
6: $\quad$ solve the free block (Step 5) with the cached factorisation of $\mathbf{K}(\tilde{G})$
7: $\quad \mathbf{w}_m \leftarrow d_m \mathbf{w}_m + \gamma_m(\mathbf{u}^{n+1} - \mathbf{u}^n)$; $\quad \mathbf{u} \leftarrow \mathbf{u}^{n+1}$
8: $\quad$ post-process (Step 8)
9: **end for**

Code: the march `nodal path`; driver `quasistatic_solid_solve!`.

Algorithm 2 One increment on the quadrature-point path

$\mathbf{u}_c \leftarrow \bar{\mathbf{u}}_c(t_{n+1})$ ▷ prescribed displacements
repeat
  $\boldsymbol{\varepsilon}_{kl} \leftarrow$ element gradient of $\mathbf{u}$ at every LGL point ▷ `_∇f!`
  $(\boldsymbol{\sigma}_{kl}, \mathbf{D}_{kl}, H^{\mathrm{tr}}_{kl}) \leftarrow$ material point$(\boldsymbol{\varepsilon}_{kl}, H^{n}_{kl}, \Delta t)$
  $\mathbf{f}^{\mathrm{int}}_a = \sum_e \sum_{kl} \omega_k \omega_l J_{kl} \nabla\psi_a \cdot \boldsymbol{\sigma}_{kl}$ ▷ weak divergence + DSS
  $\mathbf{r}_f \leftarrow \mathbf{f}^{\mathrm{ext}}_f - \mathbf{f}^{\mathrm{int}}_f$
  $\mathbf{K}_T \leftarrow \sum_e \int \mathbf{B}^{\mathsf{T}} \mathbf{D}_{kl} \mathbf{B}$; $\mathbf{u}_f \leftarrow \mathbf{u}_f + \mathbf{K}^{-1}_{T,ff} \mathbf{r}_f$
until $\|\mathbf{r}_f\| \le 10^{-10} \max(\|\mathbf{f}^{\mathrm{ext}}\|, \|\mathbf{f}^{\mathrm{int}}\|)$
$H^{n+1} \leftarrow H^{\mathrm{tr}}$ ▷ accept the history

Step 7 — the general path: quadrature-point history and Newton

For a nonlinear material (or with `:solid_history => "quadrature"`) the history H lives at every quadrature point and each increment solves $\mathbf{R}(\mathbf{u}) = \mathbf{f}^{\mathrm{int}}(\mathbf{u}; H^n) - \mathbf{f}^{\mathrm{ext}} = \mathbf{0}$ by Newton's method with the consistent tangent: With the viscoelastic model this path reproduces (24) to 10^{-10}; with J_2 plasticity it converges in two to three iterations per increment.

Code: `quadrature path`; strain `solid_qp_strains!` (with `_∇f!`); internal force `solid_internal_force!` (the contraction of `_expansion_visc!`, then `DSS_rhs!`).

Step 8 — post-processing and output

Nodal stresses (and history fields) are the mass-weighted averages of the element values at shared nodes: element gradients by `_∇f!`, summed by `DSS_rhs!`, divided by the lumped mass M_a — the SEM nodal projection. Errors are measured in the same LGL norm, $\|\boldsymbol{e}\|^2 = \sum_a M_a |\boldsymbol{e}_a|^2$. Fields go to VTK through Jexpresso's `write_output` (with the displacement also as a vector, `:vtk_vectors`, for ParaView's *Warp By Vector*); `solid_history.csv` logs every step; `:lplot_deformed` draws the deformed grid.

Code: `solid_nodal_gradient!`, `solid_qp_to_nodes!`; `write_vtk` (its `:vtk_vectors` option); `plot_deformed_mesh`.

5 Initialisation

The run starts from the undeformed, unstressed state with no history: $\mathbf{u} = \mathbf{0}$, $\mathbf{w}_m = \mathbf{0}$ (or $H = 0$: $\boldsymbol{q}_m = \boldsymbol{e}^n = \mathbf{0}$, $\boldsymbol{e}^p = \mathbf{0}$, $\bar{e}^p = 0$). `initialize` declares the state fields (`ux, uy, sxx, syy, sxy, szz`, plus any history fields to plot) and zeroes them; the solver refuses a non-zero initial displacement, because the history that produced it would not be known.

Two load conventions, chosen by `:linitial_elastic_solve`:

- **true** (default): the pressure is on at $t = 0^+$. One step with $\Delta t = 0$ ($d_m = \gamma_m = 1$, $\tilde{G} = G \sum \mu$) gives the instantaneous, glassy response, $u_A(0^+) = 0.0381333$ (Lamé); the creep follows. This is the problem the closed form solves.
- **false**: start unloaded; the full pressure acts from the first increment. Through (10) that is a load ramped over $[0, \Delta t]$ — what FEAP does when a load is switched on at step 1, and why the paper's Fig. 5.2 reads $u_A = 0.058$ at $t = 1$ where the closed form gives 0.073.

Code: `initialize`; the zero-state check and the instantaneous step `quasistatic_solid.jl`; the switch `:linitial_elastic_solve`.

6 Results and verification

6.1 Viscoelastic cylinder ($N = 4$, 4×8 elements)

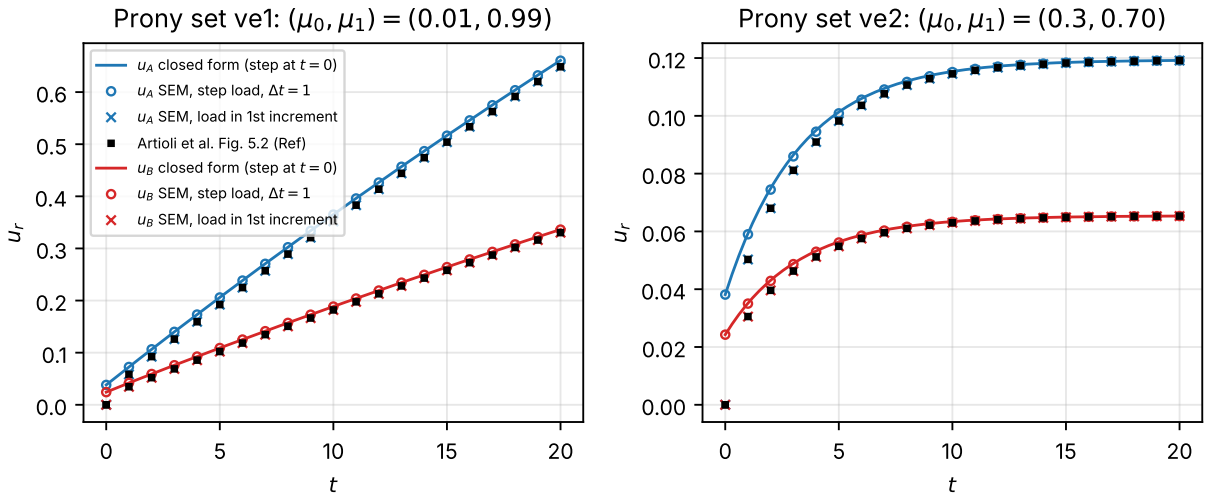


Figure 2: Radial displacement at A and B. Lines: closed form (Heaviside load). Circles: SEM with the load on at $t = 0^+$ and $\Delta t = 1$ — on the closed form within the second-order time error. Crosses: SEM with the load applied in the first increment (FEAP's convention) — on the paper's Fig. 5.2 points (squares, read from the vector graphics of the arXiv PDF).

check	result	reference
$t = 0^+$, u_A	0.0381333412	Lamé 0.0381333333
spatial error alone, $N = 2, 3, 4, 5, 6$	2.3×10^{-3}, 2.5×10^{-6}, 6.0×10^{-7}, 4.1×10^{-9}, 1.0×10^{-9}	time-discrete reference
time error, $\Delta t = 1 \to \frac{1}{8}$, exponential (ve1/ve2)	rates 1.98–2.04 / 1.97–2.00	closed form
time error, midpoint (ve1) / backward Euler (ve2)	rates 2.00–2.01 / 0.91–0.98	closed form
quadrature-point path vs nodal path	5×10^{-13} (relative)	round-off
paper Fig. 5.2, $\max\lvert u - u_{\text{paper}}\rvert$ (ve1 / ve2)	1.1×10^{-4} / 2.4×10^{-5}	figure resolution

Table 2: Verification of the viscoelastic case (`test/test_viscoelastic_cylinder.jl`, 23/23). Errors are relative to $\max u_A$ unless stated.

6.2 Elastic–perfectly plastic cylinder (the history at the quadrature points)

The same quarter cylinder, J_2 with the material of Artioli et al. §4.3 ($E = 7000$, $\nu = 0.3$, $\sigma_{y0} = 24.3$, no hardening), $N = 4$, 8×8 elements, its bore displaced radially to $\delta = 0.05$ in 50 increments. Up to first yield the solution is Lamé's ($p_e = 10.5047$); with the whole wall flowing, the pressure levels off at the exact limit $p_L = \frac{2}{\sqrt{3}}\sigma_{y0} \ln \frac{b}{a} = 19.4492$ (Hill 1950).

6.3 Seeing the solid deform

Both cases write the displacement also as a vector `u` (`:vtk_vectors`). In ParaView: open `simulation.pvd`, apply *Warp By Vector* with vector `u` (scale 1 for the viscoelastic cylinder, ∼10

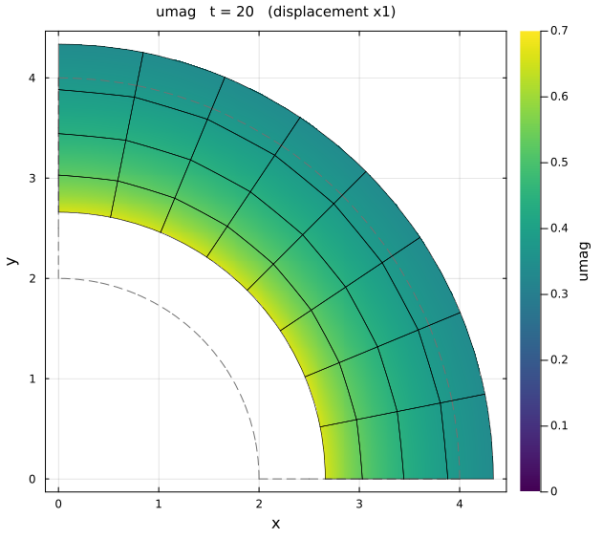


Figure 3: The deformed grid at $t = 20$ (ve1), true scale, coloured by $|\boldsymbol{u}|$, over the undeformed outline (dashed): the bore has crept from $r = 2$ to 2.66. Drawn by `plot_deformed_mesh` (`:lplot_deformed`).

check	result
J_2 point, cyclic uniaxial strain, isotropic + kinematic hardening, vs exact	1.3×10^{-16}
consistent tangent vs finite differences	$\leq 1.7\times10^{-11}$
bore pressure before first yield vs Lamé	1.9×10^{-7}, $\bar{e}^p = 0$
bore pressure vs 1D radial reference, all increments	$1.3\times10^{-4}\,p_L$
$p(\delta = 0.05)$	19.44145 (reference 19.44143, $p_L = 19.44917$)
Newton iterations per increment	3.8 on average, at most 5

Table 3: Verification of the plasticity case (`test/test_plastic_cylinder.jl`, 16/16).

for the plastic one), colour by any field (`szz`, `svm`, `ep_bar`, ...) and press *Play*. Without ParaView, `:lplot_deformed => true` writes `deformed-<field>-it<k>.png` and `deformed.gif` (Figs. 3, 4).

Code: `test/test_viscoelastic_cylinder.jl`, `test/test_plastic_cylinder.jl`.

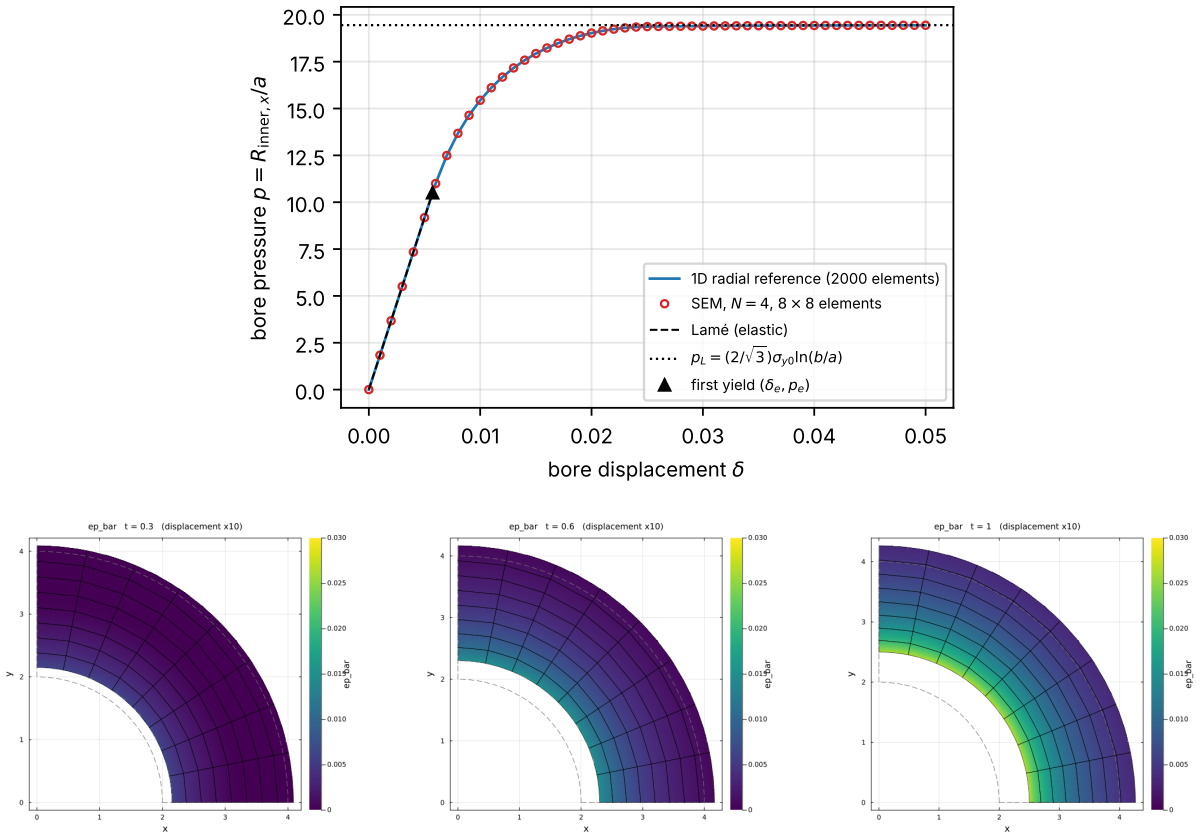


Figure 4: Top: bore pressure against bore displacement — SEM, the 1D radial reference (same constitutive routine and increments), Lamé and the exact limit. Bottom: the deformed grid (×10) coloured by the accumulated plastic strain $\bar{e}^p$, a quadrature-point history value projected onto the nodes, at $t = 0.3$, 0.6 and 1: the plastic zone spreads from the bore until the whole wall has yielded.

References

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